D · Teacher guide, alignment and marking

Students read the topic notes and study independently, without prior concept teaching or VL operation. Collect or scan first responses before giving the complete answer version. Correct in another colour without erasing originals.

Target the same 15–20 minutes for both groups, preferably 20: notes 4, fair comparison 5, measurement/analysis 8, checking 3. Pilot feasibility and record actual time, omissions and support. Do not remove core tasks to meet time. Clarify terms, scales and response procedures; document extra conceptual support.

The VL remains inquiry-first: learning points/reflection after inquiry submission, PDF after reflection submission. Paper concept-first sequencing does not change this.

Item alignment and acceptable responses

1.1 · iv,dv,cv · 4 marks

Identify the variables

Type / shared skill: Multiple checkboxes / classify experimental factors and fair-comparison controls.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

1.2 · assumptions · 1 marks

What are the assumptions of this investigation?

Type / shared skill: Multiple checkboxes / identify assumptions supporting a valid comparison.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

1.3 · controlPlan · 1 marks

If you want to find out whether antibiotics（抗生素） affect bacterial growth, how would you design a control group?

Type / shared skill: Open design / plan the matching carrier baseline without antibiotic.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

1.4 · plannedReplicates,plannedReplicateReason · 1 marks

There are 4 test samples: X, Y, Z and the control. How many tests do you think each sample needs in total? (Include the first test.) Why would you plan it this way?

Type / shared skill: Number choice and open reason / plan independent repeats, variation and feasibility; do not score the number itself.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

2.1 · measurements · 2 marks

Visible zone outside the disc? / Total clear-zone（清晰區） diameter

Type / shared skill: Presence classification and numeric readings / observe and measure total diameter including the disc.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

2.2 · means · 1 marks

Mean total diameter calculation

Type / shared skill: Numeric calculation / process the learner’s own three readings.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

2.3 · zoneMeaning · 1 marks

What does the appearance of a clear zone（清晰區） indicate?

Type / shared skill: Single choice / distinguish inhibited visible growth from complete killing.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

2.4 · zoneDiameterMeaning · 1 marks

What does the diameter of a clear zone（清晰區） represent?

Type / shared skill: Single choice / interpret diameter extent and diffusion factors.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

2.5 · effectiveSamples,rank1–4 · 1 marks

Draw a conclusion from your results:

Type / shared skill: Inline choices / identify effective samples and rank by own means.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

2.6 · repeatPurpose · 1 marks

Why do experiments need to be repeated?

Type / shared skill: Single choice / understand repeats, random error and reliability.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

2.7 · repeatVariation · 1 marks

If repeated results differ greatly, what should you do, and why?

Type / shared skill: Single choice / evaluate causes of variation without deleting disagreeing data.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

2.8 · controlPurpose · 1 marks

When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?

Type / shared skill: Single choice / interpret the baseline for non-antibiotic factors.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

3 · reflection · 8 marks

Choose an antibiotic（抗生素） sample from your experimental results that can inhibit（抑制） bacterial growth, and cite readings to support your judgement. Suppose this bacterial population initially contains a small number of bacteria resistant（具抗藥性的） to that antibiotic（抗生素）. How might its effectiveness change with continued use? Use the learning points to explain how bacterial survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵） cause this change.

Type / shared skill: Integrated response / support a chosen sample with readings and explain resistance selection causally.

Native questions/hints; repeats combine the number and reason prompts, samples use checkboxes, unused written ranks use a slash, and the final question substitutes “topic notes”.

No paper response opportunity: initial historical observation, initial predictions/reasons, independent setup drawing, aseptic handling, spreading, disc placement or incubation. Mark not applicable, not zero; paper responses do not demonstrate practical performance. The current VL has no bar chart or disc-content extension; neither is restored.

Full answers and numeric checking

See the complete answer worksheet for every prompt, hint, choice and diagram with answers. Additional acceptable responses follow.

Use a matching carrier-only disc without antibiotic（抗生素） to check carrier/disc effects, with other conditions the same. Call it a water control only if every carrier is water.

Use readings for your chosen sample to support growth inhibition（抑制）. Explain how pre-existing resistance（抗藥性） affects survival and reproduction（繁殖）, how resistance traits（抗藥性特徵） are inherited（遺傳）, and how the resistant（具抗藥性的） proportion may increase and antibiotic（抗生素） effectiveness may decline. This is a hypothetical extension, not evidence that resistance（抗藥性） arose in this experiment.

C is the same carrier without antibiotic; call it a water control only if water is the carrier. Repeat reasons connect independent plates to consistency/variation; choosing three itself earns no mark.

plate-1: X 18.7 mm, Y 0 mm, Z 26.1 mm, C 0 mm

plate-2: X 18 mm, Y 0 mm, Z 27.8 mm, C 0 mm

plate-3: X 18.9 mm, Y 0 mm, Z 27.9 mm, C 0 mm

X and Z have zones; Y and C have none and are recorded as 0. Calculate means from the three printed plates. MC keys: B, C, C, D, C. Select X and Z; fixed-data order Z > X > / > /. Judge means/order from valid own readings; accept either order for ties.

External-zone tolerance ±1 mm; absence requires the correct category and exactly 0. Check means from own three readings within ±0.1 mm, avoiding repeated reading penalties. Plates use existing seeds 1001/123456789/4294967 and generateResult, not identical outcomes for every VL run. Clear zones and 6 mm discs are drawn at 1:1 print size, without an attached ruler. Use a physical millimetre ruler. Print at 100% / actual size, never fit to page, and check the 20 mm calibration line first.

Printed model means: X 18.5 mm, Y 0.0 mm, Z 27.3 mm, C 0.0 mm

Marks, omissions and reflection

WS checking totals 24: variables 4, assumptions 1, control 1, repeat reason 1, readings 2, means 1, five MCs 5, conclusion 1, reflection 8. It is not equivalent to VL29; do not directly compare activity totals. Outcomes need researcher-selected common assessments; no test or validity evidence is invented here.

Sample/readings: 2 for appropriate sample and supporting readings; 1 for one; 0 for neither.

Existing variation/selection: 2 for pre-existing resistance and differential survival; 1 for one; 0 for neither or purposeful acquisition.

Reproduction/inheritance: 2 for both causal links; 1 for one; 0 for neither.

Proportion/effect: 2 for rising resistant proportion and possible reduced effect; 1 for one; 0 for neither.

Assess concepts and scientific causal links, not keyword counts. Accept equivalent answers using the chosen effective sample and reasonable readings; require no extra prescribed comparison, limitations or hypothesis revision. An omitted applicable response may earn 0 after marking; unmarked stays blank. Inapplicable tasks are excluded from gates.

Other criteria: IV/DV 1 each for exact sets; CVs 0.4 each for five correct controls, with 0 for inappropriate extras. Assumptions 1 for the exact three correct choices, otherwise0. Control/repeat reasons 0/1 for carrier/shared conditions or independent variation. Twelve readings 2/12 each for tolerance and category; four own-data means 0.25 each. MCs 0/1 each. Conclusion: samples0.5 and own-mean order0.5.

VL and Excel versions

analysisVersion2 task scope remains unchanged. New inquiries have rubricVersion3/taskVersion VL4-multiselect-5: reflection rises from 2 to 8; other 21 unchanged; total 29. Six manual fields: observation2, hypothesis1, repeats1, control1, design2, reflection8. R3 accepts only integers within bounds; blank=pending, 0=marked zero. Formulas reject invalid pasted values too. All applicable manual fields plus submitted reflection are required.

R2 remains23, earlier legacy32, with historical criteria and half-mark rules. Mixed exports use separate sheets and explicit assessment metadata. Original/first/latest responses, research IDs, images, events and cloud queues are retained. Manual marks live in saved Excel files; old downloads do not update—export again.

Scientific limits: no visible growth does not prove all dead. Different antibiotics diffuse differently; largest zone is not best clinical treatment, and no clinical breakpoint is supplied. Rising resistance proportion may affect effectiveness without complete takeover. Fleming did not study MRSA.

Historical photo: CBS News · The discovery of penicillin, relabelled from the source image. Figure A comes from the VL learning points; it is explanatory, separate from page 3’s measurement views printed at actual size.